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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

20EE501/17EE501/18EE501 – LINEAR INTEGRATED CIRCUITS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat block diagram explain different stages of an Op-Amp.	06	L*2	1	1
	b) Design a capacitor coupled voltage follower using a 741 op-amp. The lower cut-off frequency for the circuit is to be 50Hz and the load resistance $R_L = 3.9k\Omega$.	07	L2	1	2,3
	c) Explain the need for frequency compensation. Explain any two methods of frequency compensation. What do you mean by overcompensation?	07	L2	2	1,2
2.	a) Explain the working and design procedure to be followed for the capacitor coupled non-inverting amplifier.	06	L2	1	1,2
	b) Design a capacitor coupled voltage follower connected to single polarity power supply of +20V. Use LM741 with $f_L = 100\text{Hz}$ and $R_L = 10k\Omega$. Estimate the input impedance.	07	L3	1	2,3
	c) Explain the circuit instability due to stray capacitance. How is the compensating circuit designed?	07	L2	2	1,2
3.	a) Explain the capacitor coupled inverting amplifier circuit. Show the design steps to be followed. How to set the lower and upper cut-off frequencies?	08	L2	1	1,2
	b) Explain the capacitor coupled difference amplifier and its design.	06	L2	1	1
	c) Explain the precautions to be observed for circuit stability. Explain the need for power supply decoupling.	06	L2	2	1
Unit – II					
4.	a) Explain the working of a non-saturating precision half wave rectifier. Discuss the advantages of a precision rectifier over an ordinary diode rectifier circuit.	07	L2	3	2,3
	b) Sketch an op-amp based voltage follower peak detector circuit and explain its working. Give the design procedures to be followed.	06	L2	3	1,2
	c) A 5 kHz, $\pm 5\text{V}$ square wave with 100Ω source resistance is to have its negative peak clamped at ground level. Using bipolar op-amp design a suitable precision clamping circuit. The tilt on the output is not to exceed 2.5%.	07	L3	3	2,3
5.	a) Explain the working of inverting Schmitt Trigger circuit and draw the waveform. Show the design steps to be followed.	06	L2	3	1,2
	b) Sketch an op-amp based astable multivibrator and explain its working. Draw the circuit input output waveforms. Give the design steps and equation of frequency of operation.	07	L2	3	1,2
	c) A $\pm 1.5\text{V}$ signal is to be sampled by a Sample and Hold circuit. The holding time is to be $700\mu\text{s}$, and the output accuracy is to be 0.3%. LM108 with $I_{B(\text{max})} = 2\text{nA}$ and the FET with 25Ω channel resistance and 300nA gate-source reverse leakage current is to be used. Draw the designed the circuit and determine minimum acquisition time.	07	L3	3	2,3

P.T.O.

20EE501/17EE501/18EE501

Supple Exam - December 2025

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|-------|--|----|----|---|-----|
| 6. a) | Design a first order high pass active filter circuit to have a cutoff frequency of 5 kHz. Use LM108 op-amp and estimate the highest frequency that can be passed. Draw the circuit and frequency response. Unity gain frequency is 1MHz. | 07 | L3 | 4 | 2,3 |
| b) | Explain the working of a two stage third order active low pass filter and explain its operation. Give the design steps and selection of stage frequencies. | 08 | L2 | 4 | 1,2 |
| c) | Define figure of merit. How to calculate mid frequency? | 05 | L1 | 4 | 1 |

Unit – III

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| 7. a) | Explain the working of precision voltage regulator. Give the detailed design steps to be followed. How to calculate the performance parameters? | 10 | L2 | 5 | 2 |
| b) | With a neat internal circuit diagram explain the working of a 723 based positive voltage regulator. A 12V, 40mA LM723 regulator is to be designed with short circuit protection. The short circuit current is to be 45mA. Calculate resistor values and IC power dissipation under normal and short circuit condition. | 10 | L3 | 5 | 1,2,3 |
| 8. a) | Explain the 555-timer based astable multivibrator circuit with a neat circuit diagram and relevant waveforms. Give the design steps to vary the frequency of oscillation. | 10 | L2 | 5 | |
| b) | A 555-timer based monostable multivibrator circuit uses a 15 V supply, determine suitable R_1 and C_1 values to give 1 ms output pulse each time the circuit is triggered. Draw the circuit diagram and explain the operation with relevant waveforms. | 10 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
